

Chaiyawat Kaewmeechai

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PROFESSIONAL SUMMARY

I am a PhD student in the Condensed Matter & Materials Physics Group (CMMP) at UCL working under Prof. Alexander Shluger since 2021. My research focuses on defect modeling in amorphous and crystalline gallium oxides using density-functional theory (DFT) and molecular dynamics (MD). Specifically, I study charge trapping behaviors and defect formation in Ga_2O_3 .

In the final year of my PhD, I am modeling non-stoichiometric gallium oxides using DFT, ab initio molecular dynamics (AIMD), and classical MD. Currently, I am generating datasets and training machine-learning force fields with MACE for amorphous non-stoichiometric GaO_x systems. I plan to submit my thesis and manuscripts by March 2025.

In 2019, I completed my BSci/MSci in Physics at Chiang Mai University in Thailand, where my interest in materials modeling began. During my studies, I worked on a project involving ternary nitride semiconductors, with my thesis focusing on the interfaces between wurtzite semiconductors. After graduation, I spent a year as a researcher, designing new non-toxic halide perovskite materials $\text{Cs}_2(\text{Ti}, \text{Zr}, \text{Hf})\text{X}_6$ for use as light absorbers in solar cells, under supervision of Assoc. Prof. A. P. Jaroenjittichai and Y. Laosiritaworn.

EDUCATION

- PhD in Condensed Matter and Materials Physics, University College London, UK (April 2021 – Present, expected submission on March 2025)
- Bachelor of Science in Physics, Chiang Mai University, Thailand (June 2013 - March 2017)
- Master of Science in Physics, Chiang Mai University, Thailand (June 2017 - August 2019)

Awards and Scholarship

- Development and Promotion for Science and Technology talents project (DPST), Thailand

RESEARCH EXPERIENCE

- Investigation of band alignment between nitride and oxide wide bandgap semiconductors
- Designing non-toxic halide perovskite materials $\text{Cs}_2(\text{Ti}, \text{Zr}, \text{Hf})\text{X}_6$
- Defect and degradation mechanism of amorphous and crystalline gallium oxides

TEACHING ASSISTANT

- Computational Physics, 2022-2023, UCL, UK
- Electromagnetic Theory, 2022-2024, UCL, UK
- Practical Physics and Computing 1 (Python), 2023-2024 UCL, UK
- Linear Algebra and Vector Calculus, 2023-2024 UCL, UK

SKILLS

C/C++, python, bash, Linux, MATLAB,
DFT codes, Quantum espresso, CP2K, ABINIT
Molecular dynamics code: LAMMPS

Link: google scholar <https://scholar.google.com/citations?user=rg3R9-IAAAAJ&hl=en&oi=ao>

Highlight publications

PhD works

- Kaewmeechai, C., Strand, J., & Shluger, A. (**Under review**). Electronic structure and properties of trapped holes in crystalline and amorphous Ga_2O_3 . *Physical Review B*

Amorphous (a) Ga_2O_3 is another phase beyond crystal that can be synthesized at low-temperature. It is also considered in several applications e.g. photodetectors, resistive random access memories. We investigated the formation of self-hole trapping comparing with crystalline phases and found that only a hole can be trapped on the precursor O sites and formed a hole polaron.

- Kaewmeechai, C., Strand, J., & Shluger, A. (2024). Role of local structure in the optical and electronic properties of oxygen vacancies in different crystal phases of Ga_2O_3 . *Physical Review Materials*, 8(9), 094603.

We demonstrated that the local structures around O vacancies in α , β , and ϵ phases contribute to different charge transition levels (CTLs) and optical absorption properties. O vacancies surrounded by tetrahedral Ga atoms exhibit deeper CTLs and higher optical absorption energies.

- Kaewmeechai, C., Strand, J., & Shluger, A. (**Submitted**) Structure and migration mechanisms of oxygen interstitial defects β - Ga_2O_3 . *Physica status solidi (b)*

Previous works

- Kaewmeechai, C., Laosiritaworn, Y., & Jaroenjittichai, A. P. (2021). First-principles study on structural stability and reaction with H_2O and O_2 of vacancy-ordered double perovskite halides: $\text{Cs}_2(\text{Ti, Zr, Hf})\text{X}_6$. *Results in Physics*, 25, 104225.

$\text{Cs}_2(\text{Ti, Zr, Hf})\text{X}_6$ is considered as a potential candidate for photoluminescence and photovoltaic application, we show that $\text{Cs}_2(\text{Ti, Zr, Hf})\text{X}_6$ are lasting after reaction with H_2O but for O_2 only Cs_2ZrCl_6 and Cs_2HfCl_6 are durable.

- Kaewmeechai, C., Laosiritaworn, Y., & Jaroenjittichai, A. P. (2020). Hybrid functional investigation of band offsets for non-polar, Ga-polar and Al-polar interfaces in GaN/AlN heterojunction. *Journal of Physics: Condensed Matter*, 33(3), 035005.

Polarization along-c axis in GaN and AlN play significant role of shifting in band offsets. We observed that the polar heterojunction between these materials shown strong dipole potential energy than the non-polar heterojunction.

Other publications

- Kaewmeechai, C., Laosiritaworn, Y., & Jaroenjittichai, A. P. (2022). Band alignment of Cs_2BX_6 double halide perovskites and TiO_2 using electron affinity rule. *Results in Physics*, 42, 106015.
- Kaewmeechai, C., Laosiritaworn, Y., & Jaroenjittichai, A. P. (2021). DFT calculation on electronic properties of vacancy-ordered double perovskites $\text{Cs}_2(\text{Ti, Zr, Hf})\text{X}_6$ and their alloys: Potential as light absorbers in solar cells. *Results in Physics*, 30, 104875.
- Kaewmeechai, C., Laosiritaworn, Y., & Jaroenjittichai, A. P. (2020). DFT band alignment of polar and nonpolar GaN/MgGeN₂, ZnO/MgGeN₂ and GaN/ZnO heterostructures for optoelectronic device design. *Applied Surface Science*, 533, 147503.

Referees

1: Prof. Alexander Shluger: a.shluger@ucl.ac.uk

2: Dr. Jack Strand: jack.strand.14@ucl.ac.uk

Alex and Jack are my PhD supervisors, guiding me in defect modeling and the generation of amorphous systems.

3. Prof. Duangmanee Wongratanaphisan, duangmanee.wong@cmu.ac.th

Duangmanee, a lecturer at Chiang Mai University, specializes in perovskite solar cells. I completed both my undergraduate and master's degrees there, and she taught me quantum mechanics and solid-state physics. We often discuss challenges and advancements in the field of perovskite solar cell development.

